



Book a Cook

Technical Documentation

Tech Stack

Frontend

- **Swift & SwiftUI:** Native iOS development
- **SwiftData:** Local data persistence for user recipes and ingredients
- **Combine:** Reactive state management

Backend & Database

- **SQLite:** Pre-bundled database with 1,400+ cookbooks and 1M+ recipes
- **Firebase Authentication:** User account management (Email, Apple Sign-In, Guest)

AI & APIs

- **OpenAI GPT-4o:** Image analysis for magazine recipe extraction and kitchen scanning
- **Groq (Llama 3.1 8B):** Fast recipe generation and AI cooking instructions
- **Deepgram Nova-2:** Audio-to-text transcription for TikTok video extraction

Payments

- **RevenueCat:** Subscription management and in-app purchases

Architecture

Book a Cook follows the MVVM (Model-View-ViewModel) architecture pattern with a modular design.

Core Modules

Module	Description
HomeView	Recipe discovery with Discover, For You, and Favorites tabs
CookbooksView	ISBN barcode scanning and cookbook management
RecipesView	TikTok video extraction and magazine recipe scanning
KitchenView	Ingredient scanning and pantry/fridge/freezer management

Data Flow

- **Cookbook Database:** Pre-bundled SQLite → CookbookDatabaseManager → Views
- **User Data:** SwiftData (local) for saved recipes, ingredients, favorites
- **TikTok Extraction:** Video URL → Deepgram (transcription) → Groq (recipe generation)
- **Image Analysis:** Camera/Photo → OpenAI GPT-4o → Structured recipe/ingredients
- **Subscriptions:** RevenueCat SDK → SubscriptionManager → UsageLimitManager

RevenueCat Implementation

Configuration

Setting	Value
Entitlement ID	Book a Cook Pro
Offering ID	Pro
Monthly Product ID	com.yanxu.bookacook.pro.monthly
Yearly Product ID	com.yanxu.bookacook.pro.yearly
Monthly Price	\$1.99
Yearly Price	\$19.99

Implementation Details

SubscriptionManager.swift — Singleton class handling all RevenueCat operations:

- **configure()** — Initializes RevenueCat SDK on app launch
- **login(userID:)** — Associates RevenueCat customer with Firebase user
- **checkSubscriptionStatus()** — Fetches and validates current entitlements
- **purchase(package:)** — Handles subscription purchase flow
- **restorePurchases()** — Restores previous purchases on new devices

Subscription Gating

UsageLimitManager.swift — Enforces free tier limits:

Feature	Free	Pro
Cookbooks stored	1	Unlimited
Recipes imported	3	Unlimited
Ingredient scans	3	Unlimited
AI recipe generations	10	Unlimited

Upgrade Triggers:

- Contextual paywall shown when user attempts action beyond free limits
- Locked cookbook rows display "Unlock" button leading to paywall
- Settings page shows "Upgrade to Pro" option for free users
- "Restore Purchases" available in Settings and Paywall